

**VIETNAM NATIONAL UNIVERSITY HO CHI MINH CITY**  
**UNIVERSITY OF INFORMATION TECHNOLOGY**  
**FACULTY OF SOFTWARE ENGINEERING**



**OBJECT-ORIENTED PROGRAMMING**  
**LAB EXERCISE 1**

## PRACTICAL CONTENT

### SUBMISSION REQUIREMENTS AND INSTRUCTIONS

#### Submission instructions:

- Submit a .PDF file named: STUDENTID\_LABX.PDF (X is the lab session number; for example: 18520412\_LAB1.PDF).
- The .PDF file must follow the template provided on Courses.
- Take screenshots of the source code and place them in the template. For each method, clearly state the input, output, and the algorithmic approach used to solve the problem.
- Upload the source code to Drive and include the sharing link in the report template.

**Note: Submissions that do not satisfy the above requirements will not be graded.**

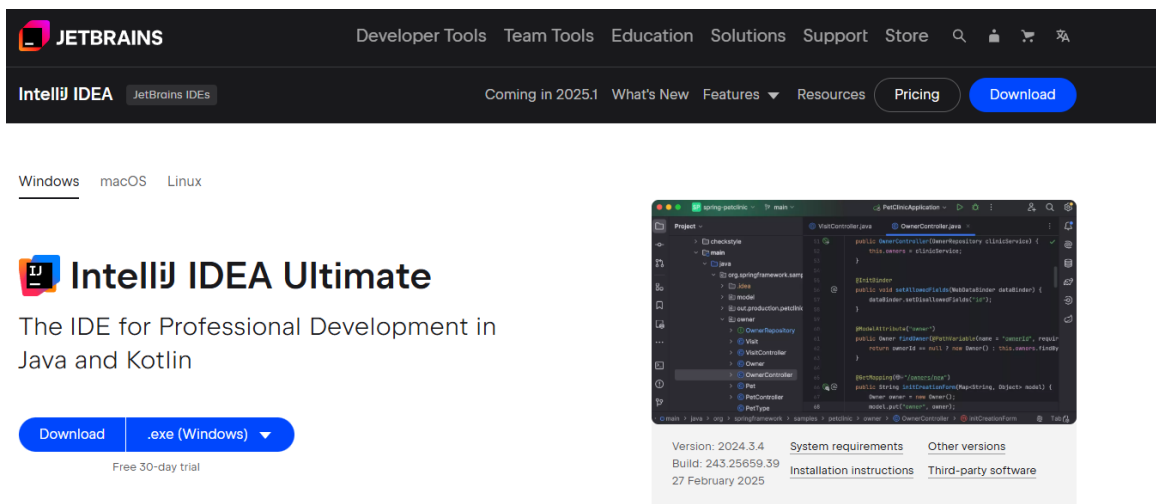
### LAB 1: JAVA BASICS

This lab introduces the basic Java programming workflow in IntelliJ IDEA and reviews variables, conditional statements, loops, strings, arrays, and simple problem-solving exercises.

#### 1. Environment Setup

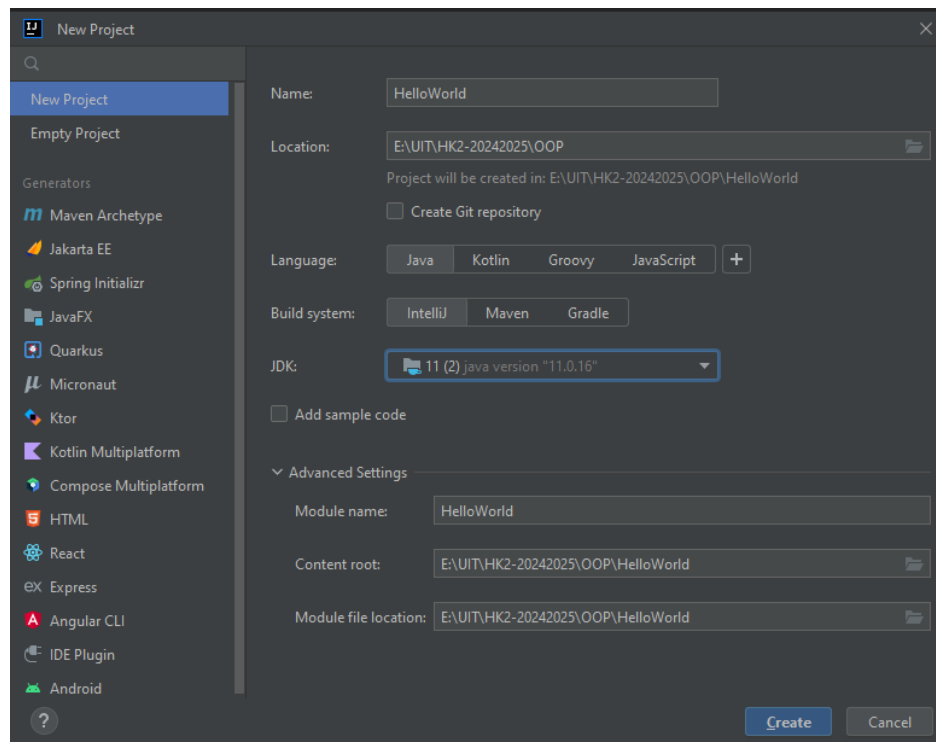
Download IntelliJ IDEA from: <https://www.jetbrains.com/idea/download/>

Create a new Java project in IntelliJ IDEA and select an available JDK.



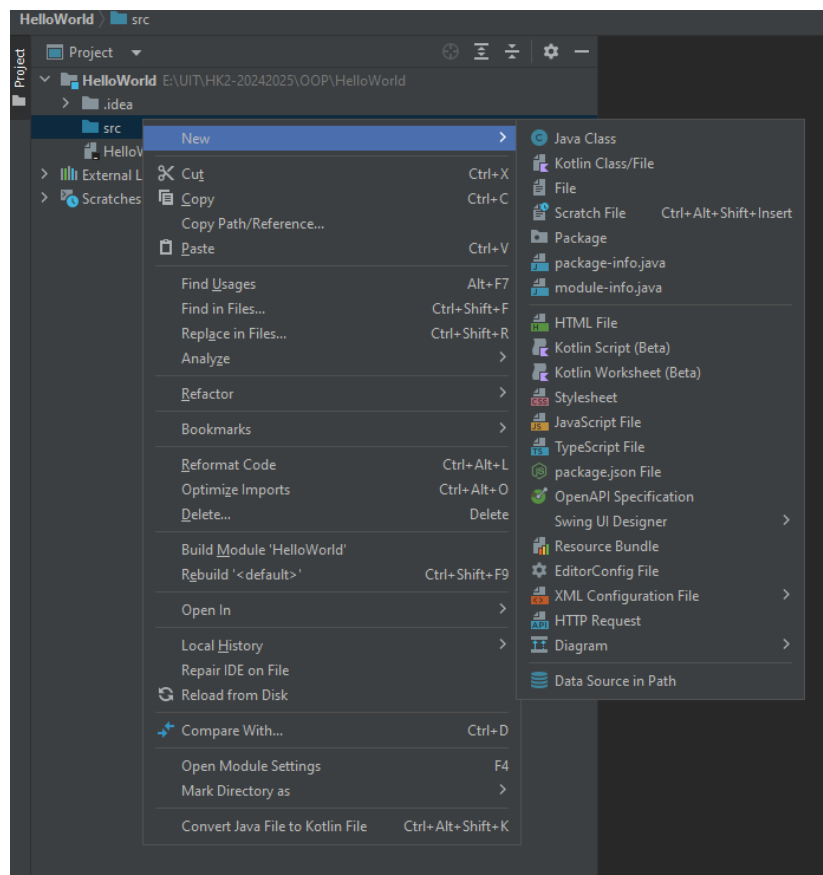
*IntelliJ IDEA download page*

## IT002 – Object-Oriented Programming

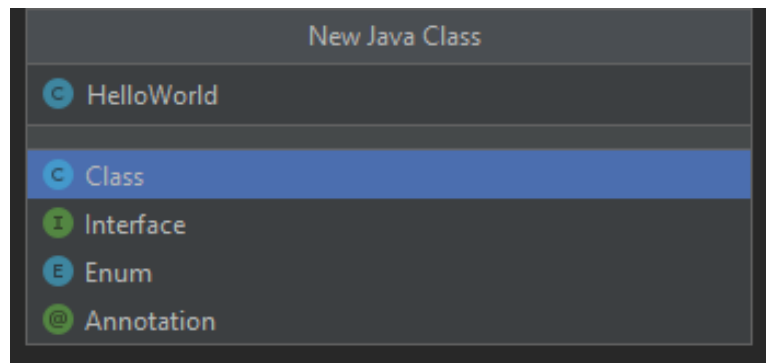


*Creating a new Java project*

Create a new Java class in the src folder by choosing New → Java Class, then enter the class name, for example HelloWorld.



*Creating a new Java class*



*Selecting the Class type*

## 2. HelloWorld Example

```
public class HelloWorld {  
    public static void main(String[] args) {  
        System.out.println("Hello world");  
    }  
}
```

Run “HelloWorld.main()”. The expected result is:

```
Hello world
```

## 3. Java Variables

Variables store data in Java. Common variable types include:

- int — integer numbers
- double — decimal numbers
- char — single characters
- String — text
- boolean — true/false values

```
int age = 25;  
double price = 19.99;  
char grade = 'A';  
String name = "John";  
boolean isStudent = true;  
  
System.out.println("Name: " + name);  
System.out.println("Age: " + age);  
System.out.println("Price: " + price);  
System.out.println("Grade: " + grade);  
System.out.println("Is a student? " + isStudent);
```

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### 4. Java If-Else Statement

The if-else statement is used to make decisions in a program.

Example: If-Else Condition

```
int number = 10;

if (number > 0) {
    System.out.println("The number is positive.");
} else if (number < 0) {
    System.out.println("The number is negative.");
} else {
    System.out.println("The number is zero.");
}
```

### 5. Java Loops

Loops help execute a block of code multiple times.

Example: For Loop

```
for (int i = 1; i <= 5; i++) {
    System.out.println("Iteration: " + i);
}
```

Example: While Loop

```
int i = 1;
while (i <= 5) {
    System.out.println("While loop iteration: " + i);
    i++;
}
```

## 6. EXERCISES

### Exercise 1: Even or Odd Numbers

Problem: Write a Java program that:

1. Takes an integer n as input.
2. Uses a loop to print all numbers from 1 to n.
3. If a number is even, print "Even"; otherwise, print "Odd".

Expected output example (if n = 10):

```
1 - Odd
2 - Even
3 - Odd
```

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```
4 - Even
5 - Odd
6 - Even
7 - Odd
8 - Even
9 - Odd
10 - Even
```

Solution:

```
import java.util.Scanner;

public class Main {
    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);

        System.out.print("Enter a number: ");
        int n = scanner.nextInt();

        for (int i = 1; i <= n; i++) {
            if (i % 2 == 0) {
                System.out.println(i + " - Even");
            } else {
                System.out.println(i + " - Odd");
            }
        }

        scanner.close();
    }
}
```

### Exercise 2: Find the Largest Number

Problem: Write a Java program that:

1. Takes three integer numbers as input.
2. Uses an if-else statement to determine the largest number.
3. Prints the largest number.

Expected input example: 10 25 30

### Exercise 3: Sum of Digits

Problem: Write a Java program that:

1. Takes an integer input from the user.

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2. Calculates the sum of its digits using a loop.
3. Prints the sum.

### **Exercise 4: Reverse a String**

Problem: Write a Java program that:

1. Takes a string as input.
2. Reverses the string using a loop.
3. Prints the reversed string.

### **Exercise 5: Multiplication Table**

Problem: Write a Java program that:

1. Takes an integer n as input.
2. Prints the multiplication table of n up to 10.

### **Exercise 6: Find an Element in an Array**

Problem: Write a Java program that:

1. Takes an integer n as input, representing the number of elements in an array.
2. Reads n integers into an array.
3. Takes another integer x as input, representing the number to search for in the array.
4. Checks whether x exists in the array.
5. If x is found, prints the index of its first occurrence. If it is not found, prints -1.

### **Exercise 7: Merge Two Sorted Arrays**

Problem: Write a Java program that:

1. Takes an integer n, representing the number of elements in the first sorted array.
2. Reads n integers into the first sorted array.
3. Takes an integer m, representing the number of elements in the second sorted array.
4. Reads m integers into the second sorted array.
5. Merges the two sorted arrays into one sorted array in ascending order.
6. Prints the merged sorted array.

## 7. BRISTLEBACK'S QUILL SPRAY DAMAGE CALCULATION

Bristleback is a very annoying hero in Dota with a well-synergized skill set and is extremely hard to kill. One of his main damage-dealing abilities is Quill Spray. This ability shoots sharp quills from Bristleback's back in all directions, making it impossible for enemy heroes to evade.

Although the initial damage of Quill Spray is not high, the ability leaves sharp quills embedded in enemy heroes. On the next cast, these existing quills will deal additional damage. Each quill exists independently, cannot be removed, and will only disappear after a certain period.

### Task

Given the times when a hero is hit by Bristleback's Quill Spray, determine the total amount of damage the hero has received.

### Notes

- If a hero gets hit by Quill Spray at the exact moment when a quill is about to disappear, the additional damage from that quill still applies before it disappears.
- For precise calculations, real numbers should be stored using the double data type.

### Input Format

1. The first line contains an integer  $n$  ( $n < 10,000$ ), the number of times the hero is hit by Quill Spray.
2. The second line contains  $n$  real numbers, representing the timestamps when the hero is hit by Quill Spray, in chronological order.
3. The third line contains a real number  $x$ , the base damage of a Quill Spray hit.
4. The fourth line contains a real number  $y$ , the additional damage dealt by each quill already embedded when Quill Spray is cast again.
5. The last line contains a real number  $z$ , the duration after which a quill disappears.

### Output Format

- Print the total damage received by the hero.

### Sample Input / Output

| Input                        | Output |
|------------------------------|--------|
| 5<br>4.9 12.6 19.1 22.9 46.2 | 580    |



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|                                                                                              |      |
|----------------------------------------------------------------------------------------------|------|
| 65<br>51<br>15.0                                                                             |      |
| 13<br>4.9 12.6 19.1 22.9 26.2 32.7 34.6 44.3<br>44.6 50.5 59.8 66.2 72.1<br>65<br>51<br>15.0 | 2120 |

### Example Calculation of Total Damage

- At second 4.9, the hero is hit by Quill Spray for the first time, receiving 65 damage and getting 1 embedded quill.
- At second 12.6, the hero is hit again, receiving 65 damage. Since there is 1 quill already embedded, they take an additional 51 damage. Now, the hero has 2 embedded quills.
- At second 19.1, the hero is hit again, receiving 65 damage. Since there are 2 embedded quills, they take an additional 102 damage. Now, the hero has 3 embedded quills.
- At second 22.9, the quill from second 4.9 has expired, leaving the hero with 2 quills. The hero is hit again, receiving 65 damage. Since there are 2 embedded quills, they take an additional 102 damage. Now, the hero has 3 embedded quills.
- At second 46.2, all quills have expired. The hero is hit again, receiving 65 damage.